CompliantWBC: Whole-Body Compliance for Heavy Humanoids via Force Latent Estimation and Residual Impedance Targets

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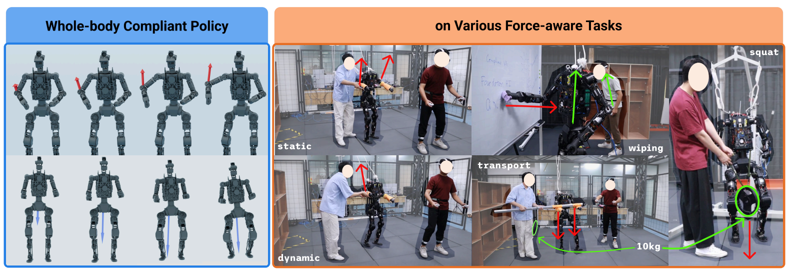

Figure 1: CompliantWBC pairs a force-aware base policy with a bounded residual on the impedance equilibrium, trained under a Phong-weighted force-origin sampler that extends compliance to the whole body. We demonstrate it on a range of force-aware tasks on a real heavy humanoid.

Abstract: Whole-body compliant control is essential for deploying heavy humanoids under high payload in human-centric environments. Most prior force-aware learning-based pipelines focus on end-effector resistance, per-link upper-body springs, or end-effector stiffness modulation, leaving *arbitrary-site perturbations on heavy platforms with lower-body engagement* largely unaddressed. We close this gap with CompliantWBC comprising: (1) A multi-site whole-body impedance *reference controller* generalizes classical Cartesian impedance to any controlled link via null-space projection and a multi-contact balancing quadratic programming (QP) and supplies per-link virtual targets that supervise an RL policy through a compliance-fidelity reward. (2) A *Force Latent Encoder*, co-trained with the base policy behind a gradient barrier and shaped by wrench reconstruction, supervised-contrastive clustering, a KL bottleneck, and temporal smoothness, yields a wrench-grounded, class-clustered latent; a bounded *Residual Policy* then edits the impedance equilibrium to absorb wrench-estimation error, preserving *local* classical-impedance passivity under fixed stiffness. (3) A *Phong-weighted force-origin sampler* with an axis-decoupled pelvis anchor induces lower-body-inclusive compliance via two interpretable parameters instead of a hand-crafted schedule. We evaluate CompliantWBC in simulation against compliant and stiff baselines and demonstrate it on a real heavy humanoid across static/dynamic force reaction, board wiping, squat under payload, and cooperative payload transport. Project website: https://compliantwbc.github.io/

Keywords: Whole-Body Compliant Control, Latent Force Modeling, Residual Policy Learning

1 Introduction

Heavy humanoid loco-manipulation requires whole-body compliance. For instance, pelvis bracing against a wall, hip translation against a pulled cart, or double-support redistribution while lifting

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cannot be expressed by upper-body springs alone. Classical operational-space methods [1, 2, 3, 4, 5, 6, 7] cover this regime analytically, but require accurate dynamics, hand-tuned task hierarchies, and contact-mode reasoning that are brittle on heavy platforms. Learning-based pipelines relax these requirements; however, the closest predecessors restrict compliance to the end-effector or to the upper-body chain [8, 9, 10, 11, 12], leaving arbitrary-site multi-contact compliance with lower-body impedance participation largely unaddressed.

In the learning-based context, the policy observes only proprioception and a motion command (the external wrench $\boldsymbol{f}_{\text{ext}}$ is unobserved on hardware) and must output torques that *yield* compliantly to $\boldsymbol{f}_{\text{ext}}$ at arbitrary contact sites while sustaining balance and command tracking. Transferring classical compliance into a learning pipeline with reinforcement learning (RL) then faces two challenges. First, the classical impedance formulation considers a single end-effector, so a principled *whole-body* heuristic must compose centroidal-momentum balance with per-link impedance at arbitrary sites and distribute the resulting wrench to a multi-contact support set. Second, training must actually exercise the whole body (not just the end-effectors) under force perturbations, and the learned policy must *encode* compliance rather than override it. Consequently, residual schemes that only edit motor commands or tracked motions [13, 14] lose the analytical controller's physical bounds. On the other hand, some whole-body trackers and teleoperation pipelines [15, 16, 17, 18, 19] expose no impedance interface during training, so compliance must be injected separately while running the full analytical pipeline at deployment inherits its runtime cost and contact-mode fragility on heavy platforms.

We address both challenges with CompliantWBC (Figure 1). For the first, we derive an analytical multi-site reference controller (§3) composing centroidal-momentum balance with per-link Cartesian impedance at arbitrary sites. Rather than cloning its torque, we use its per-link virtual targets to supervise a base RL policy through a compliance-fidelity reward [12, 20], so the deployed policy avoids the QP's runtime cost and contact-mode fragility. For the second, a Phong-weighted force-origin sampler (§4.2) drives perturbations across the whole body—including pelvis and lower-body sites that upper-body-only curricula never excite—so the policy *encodes* compliance rather than overriding it. A variational *Force Latent Encoder*, jointly trained with the base policy behind a gradient barrier from PPO and shaped by wrench reconstruction, contrastive clustering [21], a KL bottleneck, and temporal smoothness, supplies a class-clustered, wrench-grounded estimate of the unobserved contact state [22]. Finally, a bounded residual on the frozen base edits the impedance *equilibrium* rather than motor commands or gains [23, 24], so its effect is upper-bounded by the fixed stiffness and the closed loop is *locally* passive by the classical Cartesian-impedance argument (§F) [1, 25]. We study these behaviors on a heavy humanoid platform (*70 kg*, versus the *35 kg* Unitree G1 [26] used in most prior learning-based compliant humanoid work; specifications in §B) to tackle a wider range of contact-rich tasks.

Our contributions can be summarized as follows: (1) An **analytical multi-site whole-body compliance reference controller** (§3, §A) composing centroidal-momentum balance with per-link Cartesian impedance at arbitrary sites, whose per-link virtual targets serve as an RL compliance-fidelity reward rather than a behavior-cloning target. (2) A **Force Latent Encoder with residual-on-equilibrium architecture** (§4.1) whose latent is grounded in privileged wrench supervision and clustered by perturbation class, and whose bounded residual edits the impedance equilibrium, preserving analytic compliance under sim-to-real adaptation. (3) A **Phong-weighted force-origin sampler with an axis-decoupled pelvis anchor** (§4.2) that induces multi-contact, lower-body-inclusive compliance, replacing a hand-crafted per-site frequency/magnitude schedule, validated on a real heavy humanoid (§5).

2 Related Work

2.1 Compliance Control for Humanoids

Classical compliance strategies, including task-space impedance and admittance control [1, 2, 3, 4, 5, 6, 7], regulate interaction forces via virtual mass–spring–damper dynamics, with operational-

Table 1: Humanoid whole-body-control landscape. *Lower-body compliance*: yields with the legs/pelvis (✓), only in a quadruped setting (quadruped), or not at all (–). *Multi-site*: where compliance is realized—end-effector only (EE), upper body (upper only), center-of-mass reference (CoM), or arbitrary body links (whole-body). *Force latent*: an explicit, policy-conditioning force/contact embedding (✓), an implicit one (implicit), or none (–). *Residual architecture*: where a learned correction is applied.

Method	Lower-body Compliance	Multi-site	Force latent	Residual architecture	Humanoid platform
FALCON [8]	–	EE	–	–	G1/Booster
GentleHumanoid [11]	–	upper only	implicit	–	G1
FACET [10]	quadruped	CoM	–	impedance ref.	Go2/G1
SoftMimic [12]	✓	whole-body	–	–	G1
CHIP [9]	–	EE	–	hindsight goal	G1
ResMimic [14]	–	–	–	on motion	G1
ASAP [13]	–	–	–	on action	G1
CompliantWBC (ours)	✓	✓	✓	on equilibrium	in-house, 70 kg

space extensions that handle multiple frames through hierarchical task-priority projection and a balancing QP that distributes the centroidal wrench to support contacts. These pipelines provide rigorous stability guarantees but require accurate dynamics, hand-crafted task hierarchies, and explicit contact-mode reasoning that is brittle on heavy humanoid platforms; time-varying stiffness additionally breaks passivity unless an explicit storage mechanism is enforced [27, 28]. Recent whole-body tracking controllers [29, 30, 31] may achieve agile loco-manipulation via motion imitation but do not expose an impedance interface, so compliance must be injected separately.

To relax these requirements, recent learning-based methods train policies to exhibit compliant behavior through reward shaping, perturbation curricula, per-link virtual springs, or end-effector stiffness modulation [8, 11, 10, 9]. Most restrict compliance to the end-effector or the upper-body chain, leaving lower-body bracing, hip translation, and double-support redistribution—the regimes that dominate heavy loco-manipulation—unaddressed. SoftMimic [12] is a notable exception that achieves genuine whole-body compliance, but tracks a single motion rather than an arbitrary configuration. A parallel line targets heavy-payload loco-manipulation with multi-policy or trajectory-optimized references [32, 33] but similarly does not expose a per-link impedance interface. Our work composes centroidal-momentum balance and per-link Cartesian impedance at arbitrary sites inside an RL loop, admitting end-effector-only and upper-body-only formulations as special cases (Table 1).

2.2 Residual Policies and Latent Force Estimation

Residual reinforcement learning [23, 34, 35] composes a learned correction on top of a hand-crafted base controller, retaining analytical stability while absorbing unmodeled dynamics. The idea of learning residuals on controller parameters with RL was pioneered by Buchli et al. [36], who optimized both reference trajectories and gain schedules on hardware. Our work operates in the same spirit but residualizes on the impedance *equilibrium* rather than on gains. Recent humanoid pipelines instantiate this residual idea differently, focusing on motor commands, on tracked motions, or on goals via hindsight relabeling [13, 14, 9]. However, none of these approaches edit a per-link impedance interface with an explicit force bound. Consequentially, the residual's effect on induced contact force is implicit rather than physically bounded. In contrast, we compose per-link impedance at arbitrary sites and place a *bounded* residual on the per-link *equilibrium* over a *frozen* base.

A complementary line learns compact latent embeddings of unobserved context from proprioceptive history, with supervised or contrastive objectives shown to prevent collapse to motion-phase memorization [37, 38, 39, 40, 21]. Recent work specializes such latents to estimate external wrenches from proprioception without dedicated force sensors [41, 42, 43]. Relative to this RMA-style lineage, we infer a wrench-grounded, class-clustered latent from history observations behind a gradient barrier, which drives a *bounded residual on the per-link impedance equilibrium*. Because the link stiffness $\boldsymbol{K}_\ell$ is held fixed, the residual's effect is upper-bounded by an explicit virtual-target shift times the stiffness, so the closed loop is *locally* passive by the classical Cartesian-impedance argument (§F).

3 Whole-Body Impedance Heuristic

A humanoid has configuration $\boldsymbol{q} \in \mathrm{SE}(3) \times \mathbb{R}^{n_j}$, generalized velocity $\boldsymbol{\nu}$, and floating-base dynamics $\boldsymbol{M}\dot{\boldsymbol{\nu}} + \boldsymbol{C}\boldsymbol{\nu} + \boldsymbol{g} = \boldsymbol{S}^\top\boldsymbol{\tau} + \sum_c \boldsymbol{J}_c^\top \boldsymbol{\lambda}_c + \sum_i \boldsymbol{J}_{\ell_i}^\top \boldsymbol{f}_{\text{ext},i}$ with support-contact reaction wrenches $\boldsymbol{\lambda}_c$ and external wrenches $\boldsymbol{f}_{\text{ext},i}$ at links ℓ_i. Classical Cartesian impedance at a single end-effector [1] prescribes $\boldsymbol{M}_d\ddot{\boldsymbol{x}} + \boldsymbol{D}_d\dot{\boldsymbol{x}} + \boldsymbol{K}_d(\boldsymbol{x} - \boldsymbol{x}_d) = \boldsymbol{f}_{\text{ext}}$, with $\boldsymbol{x} \in \mathbb{R}^3$. We extend this to a hierarchy of frames coupled by balance.

Reference controller (summary). For controlled links $\mathcal{L}$ (hands, elbows, knees, pelvis, torso) and support contacts $\mathcal{C}_s$, the analytical reference controller composes (i) a centroidal-momentum balance task whose wrench is distributed to the support contacts by a balancing QP over friction cones and centers of pressure [5], (ii) per-link Cartesian impedance at arbitrary sites projected into the balance null space [3], and (iii) a joint-space posture task. It produces a reference torque $\boldsymbol{\tau}_{\mathrm{c}}$ and per-link virtual targets $\boldsymbol{x}^{\mathrm{c}}_{\ell,d}$. We give the full controller, including the energy-tank passivity treatment for time-varying stiffness, in §A, and use it strictly as a *reference*: of its outputs, *only* the per-link virtual targets $\boldsymbol{x}^{\mathrm{c}}_{\ell,d}$ enter the policy reward (§4). The QP-distributed torque $\boldsymbol{\tau}_{\mathrm{c}}$ is never cloned or rewarded, so the deployed policy inherits neither the QP's runtime cost nor its contact-mode fragility on heavy platforms. Admitting knees and the pelvis as controlled and support sites is the mechanism by which the heuristic extends two-footed, end-effector-only settings to multi-contact, lower-body-inclusive postures.

Force-aware virtual targets. The single signal carried into training is the per-link virtual target. For a perturbed link $\ell \in \mathcal{L}$ under an external wrench $\boldsymbol{f}_{\text{ext}}$ applied at body site p,

$$\boldsymbol{x}_{\ell,d} = \boldsymbol{x}^{\text{ref}}_\ell(t) + \boldsymbol{K}_\ell^{-1}\,\boldsymbol{f}_\ell(t), \qquad \boldsymbol{f}_\ell = \mathrm{Ad}^{-\top}_{p\to\ell}\,\boldsymbol{f}_{\text{ext}}, \tag{1}$$

where $\boldsymbol{f}_\ell$ is the external wrench transported to link ℓ's frame (equal to $\boldsymbol{f}_{\text{ext}}$ when applied at ℓ) and $\boldsymbol{K}_\ell^{-1}$ is the anisotropic Cartesian compliance, so the correction has units of displacement. This generalizes GentleHumanoid's scalar spring to arbitrary sites and anisotropic stiffness: restricting $\mathcal{L}$ to upper-body links with $\boldsymbol{K}_\ell = K_p\boldsymbol{I}_3$, $K_p \sim \mathcal{U}(5, 250)$ N/m and $\boldsymbol{D}_\ell = 2\sqrt{m_v K_p}\boldsymbol{I}_3$ ($m_v = 0.1$ kg) recovers GentleHumanoid's upper-body Cartesian-spring response [11] (up to its motion-data-driven guiding contacts), and restricting the perturbed set to the end-effectors reduces to FALCON's end-effector perturbation model [8] as an analytical special case. Time-varying $\boldsymbol{K}_\ell$ would break passivity [28]; we hold $\boldsymbol{K}_\ell$ fixed (§4.1), which makes the closed loop locally passive by the classical Cartesian-impedance argument (§F), and defer the energy-tank treatment of the general time-varying case to §A.

4 Method: Force-Aware Base Policy and Residual on Impedance Targets

4.1 Pipeline Architecture

The analytical controller of §3 assumes rigid contacts, perfect torque control, and accurate dynamics, and assumes the external wrench $\boldsymbol{f}_{\text{ext}}$ is observed; none holds on a heavy humanoid. We therefore train a two-stage pipeline (Figure 2): (i) a *force-aware base policy* that jointly trains with a Force Latent Encoder to induce compliance behavior, and (ii) a *residual policy* on top of the frozen base that compensates for the discrepancy between the base's wrench estimate and the true external wrench.

Base policy. We first train a force-aware whole-body policy $\pi_{\text{base}}(\boldsymbol{o}, \boldsymbol{c}, \boldsymbol{z}_F) \to \boldsymbol{\tau}$ that takes as input a proprioceptive observation $\boldsymbol{o}$, a motion command $\boldsymbol{c}$, and the latent wrench embedding $\boldsymbol{z}_F$ produced by the Force Latent Encoder, and outputs the joint torque. The policy is trained end-to-end with PPO [44] against a compliance-fidelity reward

$$r_t = r_{\text{track}} \;-\; w_c\,\|\boldsymbol{x}_\ell - \boldsymbol{x}^{\mathrm{c}}_{\ell,d}\|^2 \;-\; w_e\,\|\boldsymbol{\tau}\|^2, \tag{2}$$

whose term $w_c\|\cdot\|^2$ drives the controlled links toward the analytical controller's virtual-target deformation $\boldsymbol{x}^{\mathrm{c}}_{\ell,d}$ (Eq. (1)) under the ground-truth perturbation. This incentivizes the policy to learn multi-site compliance without inheriting the balancing QP's runtime cost at deployment. The full reward formulations (Table 3) and the two-stage training loop (Algorithm 1) are given in §B.

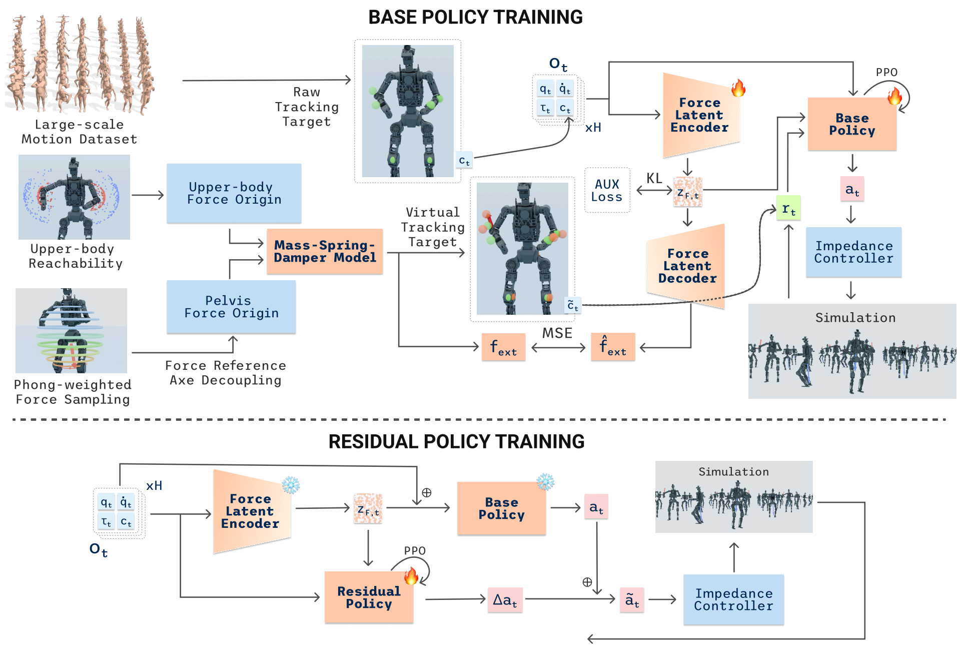

Figure 2: Two-stage training of COMPLIANTWBC: (i) a base policy jointly trained with the Force Latent Encoder to induce whole-body compliance; (ii) a bounded residual on the impedance equilibrium that compensates for physics discrepancies.

Force Latent Encoder follows a variational encoder architecture that maps an observation history window (proprioceptions, velocities, last torques, and motion command) to a force latent $\boldsymbol{z}_F \in \mathbb{R}^{d_F}$. The encoder is paired with two auxiliary heads — a *wrench decoder* ψ that regresses the per-site external wrench on all controlled links, and a *projection head* g on the unit sphere dedicated to the contrastive loss. The encoder is shaped *only* by the auxiliary objective in Eq. (3) while the posterior mean is detached before it is fed to π_{base} and π_{res}. This forms a *gradient barrier* preventing PPO gradients from flowing through the encoder, so that the latent cannot be co-opted away from its wrench-grounded objective toward reward-hacking features.

The auxiliary objective $\mathcal{L}_\phi$ combines four terms: a per-site wrench-reconstruction loss (the main term) grounding $\boldsymbol{z}_F$ in the true contact wrench through ψ; a supervised contrastive loss [21] over perturbation-class labels (which limb, which direction); a KL/VIB bottleneck [45] suppressing dimensions that carry neither wrench- nor class-relevant information; and a temporal-smoothness term on consecutive posterior means:

$$\mathcal{L}_\phi = \lambda_w \mathcal{L}_{\text{wrench}} + \lambda_s \mathcal{L}_{\text{supcon}} + \lambda_k \mathcal{L}_{\text{kl}} + \lambda_m \mathcal{L}_{\text{smooth}}. \tag{3}$$

Together they make $\boldsymbol{z}_F$ a wrench-grounded, class-clustered, low-bandwidth embedding of the privileged contact state (full forms in §B).

Residual policy is a small MLP added on top of the frozen base policy and frozen Force Latent Encoder. The residual reads $(\boldsymbol{o}_t, \boldsymbol{c}_t, \boldsymbol{z}_F, \hat{\boldsymbol{f}}_{\text{ext}})$, with the frozen wrench estimate $\hat{\boldsymbol{f}}_{\text{ext}} = \psi(\boldsymbol{z}_F)$, and outputs a bounded delta on the impedance equilibrium:

$$\tilde{\boldsymbol{x}}_{\ell,d} = \boldsymbol{x}_{\ell,d} + \epsilon_x \tanh(\Delta \boldsymbol{x}_{\ell,d}), \tag{4}$$

with positional bound $\epsilon_x = 5$ cm. When the frozen wrench estimate $\hat{\boldsymbol{f}}_{\text{ext}} = \psi(\boldsymbol{z}_F)$ is biased or noisy with respect to the true external wrench, the residual edits the impedance *equilibrium* to absorb the discrepancy without changing the base policy and without modulating gains. Editing the equilibrium rather than the torque preserves the base controller's analytic compliance: a 5 cm shift in $\boldsymbol{x}_{\ell,d}$ induces at most $\boldsymbol{K}_\ell \cdot 5$ cm of force, an explicit physical bound the residual cannot violate. We train the residual in a separate stage rather than jointly with the base: under a single PPO loss the residual would absorb compliance the base could itself learn, turning its bound into a constraint on the joint system rather than an interpretable discrepancy compensator. More details in §B.

4.2 Compliant virtual targets via external-force sampling

Phong-weighted sampling. For perturbed upper-body links (wrists, elbows) we sample force origins from a kinematic reachability cloud and drag the link toward them [11]. For the pelvis—which endures much larger forces over a limited range of motion—we instead draw a force origin from a continuous *Phong-weighted* distribution inside a ball of radius $R_a = 0.5$ m centered on the pelvis, with direction $\hat{\boldsymbol{n}}$ following a Phong lobe [46] concentrated on the downward axis $-\hat{\boldsymbol{e}}_z$ with exponent $n = 2$:

$$p(\hat{\boldsymbol{n}}) = (1-\epsilon)\,\frac{n+1}{2\pi}\,\max\!\big(-\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{e}}_z,\,0\big)^n + \epsilon\,\frac{1}{4\pi}, \tag{5}$$

with isotropic mixing weight $\epsilon = 0.1$, sampled in closed form by inverse CDF (§C) to give the offset $\boldsymbol{\delta} = r(\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ -\cos\theta)$. The downward bias emphasizes gravity on a carried load while lateral and upward samples cover reaching and lifting. Unlike a fixed cloud the distribution is gap-free, needs no forward-kinematics precomputation, and exposes its bias through two interpretable parameters (n, ϵ) (Figure 3).

Axis-decoupled pelvis anchor and compliant height target. Anchoring the offset $\boldsymbol{\delta}$ rigidly to the moving pelvis yields a self-chasing constant load, whereas a world-fixed anchor lets the force grow without bound as the robot walks away. We instead decouple the anchor by axis: the horizontal component co-moves with the floating base (leaving locomotion a bounded, body-relative load), while the vertical component—the axis we want the policy to yield along—is pinned to the *commanded* base height h^*, giving the pelvis force-origin anchor $\boldsymbol{a}_{\text{pelvis}}$:

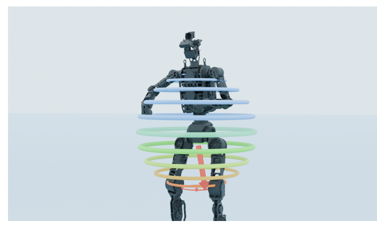

Figure 3: Phong-weighted sampling generates downward-biased force origins, decoupled by axis.

$$\boldsymbol{a}_{\text{pelvis}}(t) = \underbrace{\boldsymbol{x}^{xy}_{\text{pelvis}}(t) + \boldsymbol{\delta}^{xy}}_{\text{base-relative (horizontal)}} + \underbrace{\big(h^* + \delta^z\big)\,\hat{\boldsymbol{e}}_z}_{\text{gravity-anchored (vertical)}}, \tag{6}$$

so the anchor height $z_a = h^* + \delta^z$ is a per-chunk gravity-aligned datum that does not sink with the robot. We then evaluate the base-height term of r_{track} against a compliance-aware target $z^*_{\text{virt}}(t) = h^* + \alpha_z(z_a(t) - h^*)$, where the gain $\alpha_z = k/(k + k_{\text{leg}})$ is set by a two-spring static balance between the pelvis-anchor stiffness k and the effective vertical leg stiffness k_{leg} (§C): a stiff anchor pulls the comfortable pose toward the load, a stiff stance resists it. Penalizing $(z_{\text{pelvis}} - z^*_{\text{virt}})^2$ thus interpolates from rigid tracking ($\alpha_z = 0$) to full anchor-following ($\alpha_z = 1$), so compliant lower-body sinking emerges without an explicit compliance offset.

5 Experiments

Evaluation goal. We ask three questions: (i) does *whole-body* compliance from multi-site (pelvis, torso, elbow, wrist) force sampling improve over aggressive tracking and end-effector-only compliance, especially for forces away from the hands? (ii) does the residual on virtual targets improve force-aware tracking without sacrificing yielding? and (iii) how does COMPLIANTWBC behave in the real world on contact-rich tasks? We address (i)–(ii) in §5.2 and (iii) in §5.3.

5.1 Settings and Metrics

We compare against two external baselines—a stiff whole-body tracker **TWIST2** [15] (nominal command $\boldsymbol{x}^{\text{ref}}$, no force-aware targets) and an embodiment-specific re-implementation of upper-body-only **GentleHumanoid** [11]—and two ablations: **Ours w/o pelvis force sampling** (upper-body perturbations only, approximating GentleHumanoid-style compliance) and **Ours w/o residual** (Stage-1 base policy with the Force Latent Encoder, no Stage-2 residual). **Ours (full)** adds the bounded residual on virtual targets (Eq. (4)) with per-link stiffness fixed by the curriculum. All variants share an identical training recipe, environment budget, and paired random seeds.

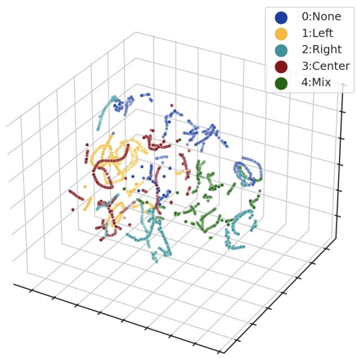

Figure 4: Force latent embedding projections via t-SNE of different force profiles across contact sites.

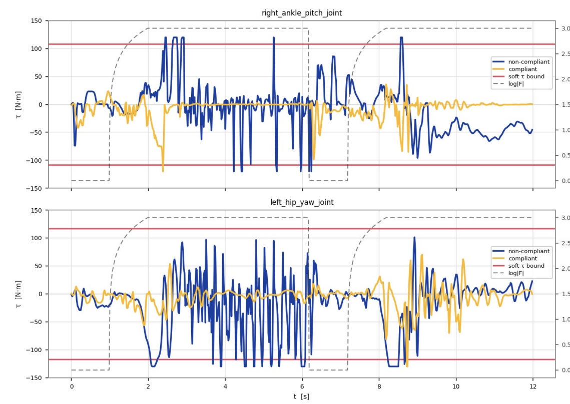

Figure 5: Compliant and Non-compliant controller reaction under pelvis perturbations.

Table 2: Performance of COMPLIANTWBC versus compliant and non-compliant baselines over 100 paired rollouts. Arrows indicate the preferred direction; bracketed columns are reported in units of 10^{-2}. $E^{\text{free}}_{\text{cmd}}/E^{\text{force}}_{\text{cmd}}$: wrist command-tracking RMSE without/with external force; ρ_τ: fraction of near-saturated joints (lower is more compliant); R_{LB}: fraction of torque deviation borne by the lower body; S: success (no fall) rate.

Controller	Tracking quality		WB compliance and robustness		
	$E^{\text{free}}_{\text{cmd}}\downarrow$ $[\times 10^{-2}]$	$E^{\text{force}}_{\text{cmd}}\downarrow$ $[\times 10^{-2}]$	$\rho_\tau\downarrow$ $[\times 10^{-2}]$	$R_{\text{LB}}\uparrow$	$S\uparrow$
TWIST2	$\mathbf{2.18} \pm \mathbf{0.31}$	$\mathbf{3.14} \pm \mathbf{0.42}$	2.74 ± 0.38	0.16 ± 0.04	0.59
GENTLEHUMANOID	5.22 ± 0.47	7.86 ± 0.61	2.12 ± 0.29	0.14 ± 0.03	0.91
OURS w/o pelvis force sampling	4.12 ± 0.38	5.89 ± 0.51	1.87 ± 0.24	0.16 ± 0.03	0.93
OURS w/o residual policy	4.29 ± 0.35	5.43 ± 0.47	1.56 ± 0.21	0.20 ± 0.03	0.94
OURS (full)	4.65 ± 0.41	6.57 ± 0.53	$\mathbf{0.93} \pm \mathbf{0.15}$	$\mathbf{0.31} \pm \mathbf{0.04}$	**0.98**

Metrics. Apart from comparing **Task success** S and **Tracking performance** $E^{\text{free}}_{\text{cmd}}$, we also report metrics that characterize the compliant response to external forces, including **Torque reaction** ρ_τ, **Lower-body participation** R_{LB}, and **Force-aware tracking** $E^{\text{force}}_{\text{cmd}}$ upon external forces. **Torque reaction** ρ_τ reflects the fraction of near-saturated joints. A stiff controller fights the force (high ρ_τ); a compliant one yields, keeping it low. **Lower-body participation** R_{LB} is the fraction of the total perturbation-induced torque deviation (relative to a matched no-perturbation rollout) borne by the lower-body joints; we treat it as a behavioral *diagnostic* of whether the policy engages the legs, not as a standalone success criterion. The metric formulations are given in the appendix.

5.2 Simulation Experiments

Protocol. Each controller runs 100 paired rollouts (a force trapezoid matching training) over a fixed site pool (wrist, elbow, torso, pelvis, hip, knee), split evenly between static (standing) and dynamic (stepping, locomotion) motions, plus a perturbation-free pass per *(controller, motion)* pair for the free-space and lower-body-participation metrics.

Results. Table 2 lays out the tracking–compliance trade-off. TWIST2 tracks best in free space but succeeds on only 59% of trials, fighting every perturbation; GENTLEHUMANOID reaches $S = 0.91$ by relaxing the upper-body limbs but shows the lowest whole-body engagement ($R_{\text{LB}} = 0.14$). OURS (FULL) sits on the favorable frontier, with the highest lower-body participation and the lowest joint saturation ρ_τ at the highest success rate, at only a modest free-tracking gap to TWIST2. The ablations isolate each component: removing pelvis sampling collapses R_{LB} toward the GENTLEHUMANOID regime, while removing the residual recovers slight tracking precision at the cost of compliance and robustness, demonstrated by worst joint saturation, lower-body participation, success rate. We provide further qualitative insight into the learned behavior of the Force Latent Encoder and the compliant response of the policy. Figure 4 shows the Force Latent Encoder organizes its embeddings into distinct per-site clusters (left/right limb, pelvis, mixed, and no-perturbation)—a force-origin-aware representation rather than a collapsed force/no-force mode, a prerequisite for

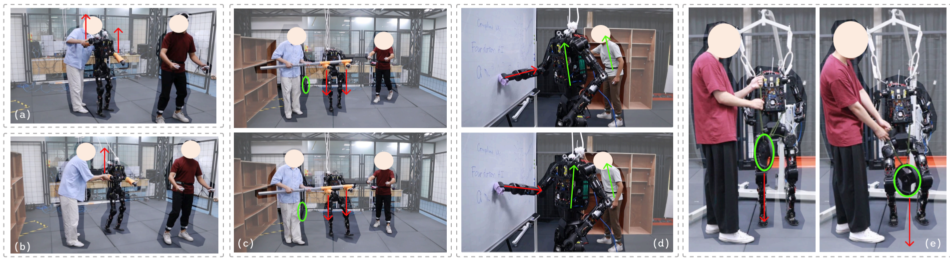

Figure 6: We deploy COMPLIANTWBC in the real world on five force-aware tasks: (a) Static Force Reaction; (b) Dynamic Force Reaction; (c) Cooperative Payload Transport; (d) Board Wiping; and (e) Squat Under Payload.

site-dependent reactions. Figure 5 contrasts compliant and non-compliant joint torques under a pelvis perturbation: the compliant policy keeps lower-body torques much smaller and stays stable through the contact, consistent with its lower ρ_τ in Table 2 and concretizing the resist-and-fail behavior behind TWIST2's low success rate.

5.3 Real-World Experiments

We deploy our trained policy on a real humanoid via teleoperation [47, 48, 49] on five tasks: *Static Force Reaction*, *Dynamic Force Reaction*, *Cooperative Payload Transport*, *Board Wiping*, and *Squat Under Payload* (Figure 6). The first two mirror the simulation perturbation protocol; the remaining three apply sustained, structured contacts that no single-site end-effector formulation can absorb, directly stressing the multi-site, whole-body assumptions of our pipeline (§4). We report these as *qualitative* demonstrations; quantitative on-robot evaluation is left to future work.

Under large pushes while tracking a dynamic walking motion (Figure 6b,c) the robot stays balanced—the regime where the stiff TWIST2 collapses in simulation ($S = 0.59$). In *Cooperative Payload Transport* the operator hands over a $100\,\mathrm{N}$ bar; the robot yields at the wrists, redistributes the load through torso and hips, and re-establishes a stable gait—the yield-then-reconcile behavior of the impedance equilibrium (Eq. (1)). *Squat Under Payload* stresses the same mechanism vertically, spreading sustained pelvis load across the chain rather than saturating the hip and knee. *Board Wiping* is the clearest whole-body case: as the operator presses a board against the robot's hand, the humanoid leans its trunk back and shifts its CoM over the support polygon—which an upper-body-only baseline cannot do, since no perturbation reaches its lower body in training. These demonstrations are qualitatively consistent with the simulation ablation.

6 Conclusion

We presented COMPLIANTWBC, a two-stage RL pipeline that defines a multi-site whole-body impedance reference controller and uses it as a compliance-fidelity reward signal. Stage 1 jointly trains the Force Latent Encoder and a force-aware base policy against this reward while stage 2 trains a bounded residual on the impedance equilibrium that compensates for the discrepancy between the frozen wrench estimator and the true external wrench, with per-link stiffness held fixed so that passivity follows from the classical Cartesian-impedance argument. We validate the result on heavy-payload loco-manipulation tasks that require lower-body and multi-contact compliance and observe compliance behavior with high lower-body participation both in simulation and in the real world.

Limitations & Future Work. Our passivity result is *local* (fixed stiffness, near the equilibrium, assuming the base realizes the analytical torque; §F), and the balancing QP (§A) can become infeasible under large simultaneous perturbations. The Force Latent Encoder is trained on simulated ground-truth wrenches, so hardware relies on proprioceptive generalization; a force–torque or current-based estimator [43, 41] would help. Finally, experiments use a single heavy platform and the real-world study is qualitative, so isolating the heaviness advantage and adding quantitative on-robot metrics remain key next steps, alongside bimanual sustained-contact manipulation and end-to-end stiffness learning under a differentiable energy-tank gate.

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A Analytical whole-body reference controller

This appendix gives the full analytical compliance controller summarized in §3. It is used *only* as a reference that generates the per-link virtual targets $\boldsymbol{x}^{\mathrm{c}}_{\ell,d}$ of Eq. (1) used in the policy reward; the reference torque $\boldsymbol{\tau}_{\mathrm{c}}$ and the energy tank below are derivation/reference quantities and are *not* executed at deployment.

A humanoid has configuration $\boldsymbol{q} \in \mathrm{SE}(3) \times \mathbb{R}^{n_j}$, generalized velocity $\boldsymbol{\nu}$, and floating-base dynamics $\boldsymbol{M}\dot{\boldsymbol{\nu}} + \boldsymbol{C}\boldsymbol{\nu} + \boldsymbol{g} = \boldsymbol{S}^\top \boldsymbol{\tau} + \sum_c \boldsymbol{J}_c^\top \boldsymbol{\lambda}_c + \sum_i \boldsymbol{J}_{\ell_i}^\top \boldsymbol{f}_{\text{ext},i}$ with support-contact reaction wrenches $\boldsymbol{\lambda}_c$ and external wrenches $\boldsymbol{f}_{\text{ext},i}$ at links ℓ_i. Classical Cartesian impedance at a single end-effector [1] prescribes $\boldsymbol{M}_d\ddot{\boldsymbol{x}} + \boldsymbol{D}_d\dot{\boldsymbol{x}} + \boldsymbol{K}_d(\boldsymbol{x} - \boldsymbol{x}_d) = \boldsymbol{f}_{\text{ext}}$; we lift this to a hierarchy of frames coupled by balance.

Reference compliant torque. For controlled links $\mathcal{L}$ (hands, elbows, knees, pelvis, torso), support contacts $\mathcal{C}_s$, and centroidal momentum $\boldsymbol{h}$, the reference torque is

$$\boldsymbol{\tau}_{\mathrm{c}} = \underbrace{\boldsymbol{J}_{\boldsymbol{h}}^\top \boldsymbol{w}_{\boldsymbol{h}}^{\text{imp}}}_{\text{centroidal balance}} + \underbrace{\sum_{\ell \in \mathcal{L}} \boldsymbol{N}_{\text{bal}}^\top \boldsymbol{J}_\ell^\top [\boldsymbol{K}_\ell(\boldsymbol{x}_{\ell,d} - \boldsymbol{x}_\ell) + \boldsymbol{D}_\ell(\dot{\boldsymbol{x}}_{\ell,d} - \dot{\boldsymbol{x}}_\ell)]}_{\text{per-link Cartesian impedance}} + \underbrace{\boldsymbol{N}_{\text{task}}^\top [\boldsymbol{K}_q(\boldsymbol{q} - \boldsymbol{q}_{\text{nom}}) + \boldsymbol{D}_q\dot{\boldsymbol{q}}]}_{\text{joint-space posture}} \tag{7}$$

where the centroidal-impedance wrench is $\boldsymbol{w}_{\boldsymbol{h}}^{\text{imp}} = \boldsymbol{K}_h(\boldsymbol{h}_d - \boldsymbol{h}) + \boldsymbol{D}_h(\dot{\boldsymbol{h}}_d - \dot{\boldsymbol{h}}) + m\boldsymbol{g}$, $\boldsymbol{N}_{\text{bal}} = \boldsymbol{I} - \boldsymbol{J}_{\boldsymbol{h}}^\top (\boldsymbol{J}_{\boldsymbol{h}} \boldsymbol{M}^{-1} \boldsymbol{J}_{\boldsymbol{h}}^\top)^{-1} \boldsymbol{J}_{\boldsymbol{h}} \boldsymbol{M}^{-1}$ is the dynamically consistent null-space projector of the balance task [3], and $\boldsymbol{N}_{\text{task}}$ projects orthogonal to both balance and per-link tasks. Each per-link Cartesian wrench is first mapped to joint torque by $\boldsymbol{J}_\ell^\top$ and then projected by $\boldsymbol{N}_{\text{bal}}^\top$, so every term acts in the n_j-dimensional generalized-force space. The desired centroidal wrench is realized by the support contacts via a balancing QP in the spirit of [5]:

$$\min_{\{\boldsymbol{\lambda}_c\}} \left\| \boldsymbol{w}_{\boldsymbol{h}}^{\text{imp}} - \textstyle\sum_c \boldsymbol{G}_c \boldsymbol{\lambda}_c - \sum_i \boldsymbol{G}_{\text{ext},i} \boldsymbol{f}_{\text{ext},i} \right\|^2 + \gamma_{\text{reg}} \textstyle\sum_c \|\boldsymbol{\lambda}_c\|^2 \ \text{ s.t. } \ \boldsymbol{\lambda}_c \in \mathcal{F}_{\mu_c},\ \mathrm{CoP}_c \in \mathcal{S}_c \tag{8}$$

with grasp maps $\boldsymbol{G}_c$, linearized friction cones $\mathcal{F}_{\mu_c}$, support polygons $\mathcal{S}_c$, and regularizer weight γ_{reg}. Admitting knees into $\mathcal{C}_s$ adds support sites and generalizes the two-footed setting to multi-contact, lower-body-involved postures.

Energy tank (general time-varying-stiffness case). Time-varying $\boldsymbol{K}_\ell(t)$ with null-space projection injects power and breaks passivity [27, 28]. For that general case one may add a scalar tank energy $T(t) \in [T_{\min}, T_{\max}]$,

$$\dot{T} = \textstyle\sum_\ell \dot{\boldsymbol{x}}_\ell^\top \boldsymbol{D}_\ell \dot{\boldsymbol{x}}_\ell - P_{\text{inject}}(\dot{\boldsymbol{K}}_\ell, \boldsymbol{N}_{\text{bal}}, \boldsymbol{N}_{\text{task}}), \tag{9}$$

filled by dissipated power and drained by non-passive control, freezing gains as $T \to T_{\min}$. In our deployed pipeline $\boldsymbol{K}_\ell$ is held fixed and the residual edits only the equilibrium, so this tank is inert at runtime and is retained only as scaffolding for the time-varying-stiffness extension (cf. §F).

B Architecture details

Base policy. 3-layer MLP, hidden width 512, trained with PPO [44]. Observation $\boldsymbol{o}$ comprises joint configurations, velocities, projected gravity, the last action, and the motion command $\boldsymbol{c}$. The latent $\boldsymbol{z}_F$ enters as an additional input.

Force Latent Encoder. The encoder takes a history window $\boldsymbol{h}_t = (\boldsymbol{q}_{t-H:t}, \boldsymbol{\nu}_{t-H:t}, \boldsymbol{\tau}_{t-H:t-1}, \boldsymbol{c}_{t-H:t})$ of $H = 16$ steps (160 ms at 100 Hz), flattens it, and passes it through a 3-layer MLP (widths 512 → 512 → 256, LayerNorm on the last hidden layer) emitting $(\boldsymbol{\mu}_t, \log \boldsymbol{\sigma}_t^2)$ over a latent of dimension $d_F = 32$. A reparameterized sample $\boldsymbol{z}_t = \boldsymbol{\mu}_t + \boldsymbol{\sigma}_t \odot \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$, drives the wrench head $\psi : \boldsymbol{z}_F \to \hat{\boldsymbol{f}}_{\text{ext}} \in \mathbb{R}^{3|\mathcal{L}|}$ on all $|\mathcal{L}|$ controlled links and the projection head $g : \boldsymbol{z}_F \to \mathbb{S}^{d_g - 1}$ with $d_g = 64$.

Auxiliary losses.

$$\mathcal{L}_{\text{wrench}} = \big\|\psi(\boldsymbol{z}_t) - \boldsymbol{f}_{\text{ext}}\big\|_2^2, \tag{10}$$

$$\mathcal{L}_{\text{supcon}} = \mathcal{L}^{\text{sup}}\big(g(\boldsymbol{z}_t),\, y_t\big), \tag{11}$$

$$\mathcal{L}_{\text{kl}} = \mathrm{D}_{\text{KL}}\big(\mathcal{N}(\boldsymbol{\mu}_t, \boldsymbol{\sigma}_t^2) \,\big\|\, \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})\big), \tag{12}$$

$$\mathcal{L}_{\text{smooth}} = \big\|\boldsymbol{\mu}_t - \boldsymbol{\mu}_{t-1}\big\|_2^2. \tag{13}$$

Weights: $(\lambda_w, \lambda_s, \lambda_k, \lambda_m) = (1.0,\, 0.3,\, 10^{-3},\, 0.1)$. $\mathcal{L}^{\text{sup}}$ is the supervised contrastive loss of [21] with perturbation-class labels y_t indicating which limb is perturbed and in which direction.

Residual policy. 3-layer MLP, hidden width 256, trained with PPO against the same reward of Eq. (2). Output is bounded as in Eq. (4) with $\epsilon_x = 5$ cm; the per-link stiffness $\boldsymbol{K}_\ell$ is held fixed by the curriculum throughout Stage 2.

C Force sampling details

Phong-weighted force-origin sampling. The inverse-CDF expressions for the Phong lobe of Eq. (5) are

$$\cos\theta = u^{1/(n+1)}, \qquad \phi \sim \mathcal{U}[0, 2\pi), \qquad r = R_a\, v^{1/3}, \tag{14}$$

with $u, v \sim \mathcal{U}[0,1]$, θ measured from $-\hat{\boldsymbol{e}}_z$. The cube-root radial warp yields uniform density in volume; with probability ϵ the direction is drawn isotropically instead.

The pelvis stiffness (k, c) is set so the spring force at maximum displacement R_a reaches a fixed fraction of the robot's nominal weight: the stiffness term yields a coherent load that grows as the policy drifts from the anchor and admits a well-defined rest position against the leg stiffness, while the damping term reproduces the velocity-dependent tug of an elastically attached inertia. Per chunk we draw hold time $T_{\text{hold}} \sim \mathcal{U}(0.2, 2.0)$ s, ramp-down $\mathcal{U}(0.25, 1.0)$ s, and upper-body stiffness $K_p \sim \mathcal{U}(5, 250)$ N/m, with upper-body type sampled from {none, single-arm, both arms}; knees and feet are never perturbed directly. The shared downward bias of Eq. (5) makes the dominant carried-load regime coherent without an explicit pelvis-to-upper-body direction-inheritance heuristic.

Force law. Upper-body anchors apply a unilateral spring that auto-ramps as the body approaches and releases [11], while the pelvis is driven by a mass–spring–damper coupling to the sampled anchor to model a heavy object held through a compliant grasp. The two laws share the same anchor-based form and are gated on link type:

$$\boldsymbol{f}_\ell(t) = \begin{cases} k\big(\boldsymbol{a}_{\text{pelvis}}(t) - \boldsymbol{x}_{\text{pelvis}}(t)\big) - c\,\dot{\boldsymbol{x}}_{\text{pelvis}}(t), & \ell = \text{pelvis}, \\ K_p\big(\boldsymbol{a}_\ell - \boldsymbol{x}_\ell(t)\big)\, \mathbf{1}[\boldsymbol{a}_\ell \text{ on active side}], & \ell \in \mathcal{L}_{\text{upper}}. \end{cases} \tag{15}$$

D Algorithm

Reward. The trained objective is the compliance-fidelity reward of Eq. (2), where $\boldsymbol{x}^{\text{c}}_{\ell,d}$ is the analytical controller's impedance-implied trajectory under the realized (ground-truth) perturbation, available in simulation. The term $w_c\|\cdot\|^2$ rewards following this compliant deformation rather than the original motion command—generalizing GentleHumanoid's [11] reference-dynamics reward to the whole body. The encoder's wrench-reconstruction MSE does *not* appear in r_t: it shapes ϕ only through $\mathcal{L}_\phi$ (Eq. (3)), behind the gradient barrier, so the policy reward never confuses "latent fits the wrench" with "robot behaves compliantly."

Hyperparameters. $N_1 = 5{\times}10^8$ environment-steps for Stage 1; $N_2 = 5{\times}10^8$ environment-steps for Stage 2; 8192 parallel environments in MjWarp (MuJoCo-based [50]); PPO with clip 0.2, discount $\gamma = 0.99$, GAE $\lambda = 0.95$; control rate 100 Hz; reward weights $w_c = 0.5$, $w_e = 10^{-3}$; force-latent auxiliary weights $(\lambda_w, \lambda_s, \lambda_k, \lambda_m)$ as in Eq. (3).

Algorithm 1 COMPLIANTWBC training pipeline

Require: Motion dataset $\mathcal{M}$, force-sampling distribution $\mathcal{C}$ (§4.2), simulator $\mathcal{S}$

Stage 1: Force-aware base policy (PPO) + Force Latent Encoder (aux).

Initialize π_{base}, encoder ϕ, wrench head ψ, projection head g randomly.
for $i = 1, \ldots, N_1$ **do**
  Sample $(m, \boldsymbol{f}_{\text{ext}}, y, \mathcal{C}_s) \sim \mathcal{M} \times \mathcal{C}$ (y: perturbation class).
  for $t = 1, \ldots, T$ **do**
    $(\boldsymbol{\mu}_t, \log \boldsymbol{\sigma}_t^2) \leftarrow \phi(\boldsymbol{q}_{t-H:t}, \boldsymbol{\nu}_{t-H:t}, \boldsymbol{\tau}_{t-H:t-1}, \boldsymbol{c}_{t-H:t})$; $\boldsymbol{z}_F \leftarrow \boldsymbol{\mu}_t$.detach().
    $\boldsymbol{\tau}_t \leftarrow \pi_{\text{base}}(\boldsymbol{o}_t, \boldsymbol{c}_t, \boldsymbol{z}_F)$; step $\mathcal{S}$.
    Roll the analytical controller on the same state to obtain $\boldsymbol{x}^{\text{c}}_{\ell,d}$; compute reward (2).
  end for
  PPO update on π_{base} [44].
  Aux update on (ϕ, ψ, g): $\nabla \mathcal{L}_\phi$ from Eq. (3) using $(\boldsymbol{f}_{\text{ext}}, y)$.
end for

Stage 2: Residual on impedance targets (PPO).

Freeze π_{base}, ϕ, ψ. Initialize the residual policy π_{res} randomly.
for $i = 1, \ldots, N_2$ **do**
  Sample $(m, \boldsymbol{f}_{\text{ext}}, \mathcal{C}_s) \sim \mathcal{M} \times \mathcal{C}$.
  for $t = 1, \ldots, T$ **do**
    $\boldsymbol{z}_F \leftarrow \phi(\cdot)$; $\hat{\boldsymbol{f}}_{\text{ext}} \leftarrow \psi(\boldsymbol{z}_F)$.
    Form analytical target $\boldsymbol{x}_{\ell,d}$ via Eq. (1) using $\hat{\boldsymbol{f}}_{\text{ext}}$.
    $\Delta \boldsymbol{x}_{\ell,d} \leftarrow \pi_{\text{res}}(\boldsymbol{o}_t, \boldsymbol{c}_t, \boldsymbol{z}_F, \hat{\boldsymbol{f}}_{\text{ext}})$.
    Compose $\tilde{\boldsymbol{x}}_{\ell,d}$ via (4); $\boldsymbol{K}_\ell$ held fixed by curriculum.
    $\boldsymbol{\tau}_t \leftarrow \pi_{\text{base}}(\boldsymbol{o}_t, \boldsymbol{c}_t, \boldsymbol{z}_F)$ executed against $(\tilde{\boldsymbol{x}}_{\ell,d}, \boldsymbol{K}_\ell)$; step $\mathcal{S}$; compute reward (2).
  end for
  PPO update on π_{res} [44].
end for
return $\pi_{\text{base}}, \phi, \psi, \pi_{\text{res}}$.

E Metrics Formulation

Task success. A trial is successful if the robot remains valid throughout the force perturbation rollout without triggering the evaluation termination conditions. In our evaluation, these terminations are restricted to fall/instability events: root linear velocity exceeding $5\,\text{m/s}$, root height dropping below $0.5\,\text{m}$, or root height rising above $1.0\,\text{m}$. Let $f_i \in \{0, 1\}$ denote whether any such termination is triggered during trial i. The success indicator is

$$s_i = \mathbf{1}[f_i = 0], \tag{16}$$

and the reported success rate is

$$S = \frac{1}{N} \sum_{i=1}^{N} s_i. \tag{17}$$

Command tracking error. For the commanded body frames $\mathcal{K}_{\text{cmd}}$, currently the left and right wrist frames, we measure Cartesian tracking error relative to the reference motion. Let $\mathcal{T}_{\text{free}}$ denote timesteps for which the external-force envelope is inactive, and let $\mathcal{T}_{\text{force}}$ denote timesteps for which it is active, i.e., the normalized applied-force envelope $a_t > 0.05$ (the trapezoid amplitude of §4.2). For a time set $\mathcal{T}$, the command tracking RMSE is

$$E_{\text{cmd}}(\mathcal{T}) = \sqrt{\frac{1}{|\mathcal{T}|\,|\mathcal{K}_{\text{cmd}}|} \sum_{t \in \mathcal{T}} \sum_{k \in \mathcal{K}_{\text{cmd}}} \left\| \boldsymbol{p}_{k,t} - \boldsymbol{p}^{\text{ref}}_{k,t} \right\|_2^2}. \tag{18}$$

We report the free-space error $E^{\text{free}}_{\text{cmd}} = E_{\text{cmd}}(\mathcal{T}_{\text{free}})$ from the no-force pass and the force-window error $E^{\text{force}}_{\text{cmd}} = E_{\text{cmd}}(\mathcal{T}_{\text{force}})$ over the active perturbation window, which quantifies tracking relaxation under external contact.

Table 3: Full reward bundle for COMPLIANTWBC (both stages), grouped by purpose with per-term weights; the compliance-fidelity term of Eq. (2) corresponds to the *Compliance* group.

Term	Form	Weight
Whole-body tracking		
`tracking_joint_dof`	$\exp\big(-0.15 \sum_j (q_j - q_{j,d})^2\big)$	$+2.0$
`tracking_joint_vel`	$\exp\big(-0.01 \sum_j (\dot{q}_j - \dot{q}_{j,d})^2\big)$	$+0.2$
`tracking_root_rotation`	$\exp(-5\,\theta_q^2)$	$+1.0$
`tracking_root_linear_vel`	$\exp(-\lVert \mathbf{v}_b - \mathbf{v}_{b,d}\rVert^2)$	$+1.0$
`tracking_root_angular_vel`	$\exp(-\lVert \boldsymbol{\omega}_b - \boldsymbol{\omega}_{b,d}\rVert^2)$	$+1.0$
`tracking_keybody_pos`	$\exp\big(-10 \sum_k \lVert \Delta_b^k - \Delta_{b,d}^k\rVert^2\big)$	$+2.0$
`tracking_keybody_pos_global`	$\exp\big(-10 \sum_k \lVert \mathbf{x}_k - \mathbf{x}_{k,d}\rVert^2\big)$	$+0.1$
Compliance — admittance force & virtual target		
`force_reward`	$\exp\big(-0.0625\,\langle \lVert \mathbf{F}_\ell^{\mathrm{app}} - \mathbf{F}_\ell^{\mathrm{exp}}\rVert\rangle_\ell^2\big)\,\mathbb{1}[\text{not exc.}]$	$+2.0$
`force_target_tracking`	$\exp\big(-25\,\langle \lVert \mathbf{x}_\ell - \tilde{\mathbf{x}}_\ell\rVert\rangle_\ell^2\big)$	$+2.0$
`force_target_vel_tracking`	$\exp\big(-\langle \lVert \dot{\mathbf{x}}_\ell - \dot{\tilde{\mathbf{x}}}_\ell\rVert\rangle_\ell^2\big)$	$+1.0$
`force_exd_penalty`	$\langle \mathbb{1}[\lVert \mathbf{F}_\ell^{\mathrm{app}}\rVert > F_{\max,\ell} \wedge \lVert \mathbf{F}_\ell^{\mathrm{app}}\rVert > \lVert \mathbf{F}_\ell^{\mathrm{exp}}\rVert + \frac{\delta}{2}]\rangle_\ell$	-6.0
`keypoint_tracking_imp`	$\exp\big(-10 \sum_{k\in\mathcal{K}} \lVert \mathbf{x}_k - \tilde{\mathbf{x}}_k\rVert^2\big),\ \tilde{\mathbf{x}}_k = \mathbf{x}_\ell^{\mathrm{virt}}$ on $\mathcal{L}$	$+2.0$
Lower-body yielding under load		
`lower_keypoint_tracking`	$\exp\big(-10 \sum_{k\in\mathcal{K}_{\mathrm{lo}}} \lVert \mathbf{x}_k - \tilde{\mathbf{x}}_{k,\mathrm{lo}}\rVert^2\big)$	$+2.0$
`tracking_compliant_root_translation_z`	$\exp\big(-5\,(z_{\mathrm{root}} - z_{\mathrm{virt}}^*)^2\big)$	$+1.0$
`squat_reward`	$\exp\big(-25\,(\Delta z - c_z \sum_\ell \lVert \mathbf{F}_\ell\rVert)^2\big),\ c_z = 3\times10^{-4}$ m/N	$+1.0$
Safety — joint, torque, contact, collision		
`alive`	$\mathbb{1}[\text{alive}]$	$+0.5$
`dof_pos_limits`	$\sum_j\big[\mathrm{ReLU}(q_j - q_j^{\max}) + \mathrm{ReLU}(q_j^{\min} - q_j)\big]$	-5.0
`joint_limit`	same form as `dof_pos_limits` (additional)	-10.0
`dof_torque_limits`	$\sum_j \mathrm{ReLU}\big(\lvert\tau_j\rvert/\tau_{j,\max} - 0.95\big)$	-1.0
`self_collisions`	$\mathbb{1}[\,\lVert \mathbf{F}^{\mathrm{self}}\rVert > 10\,\mathrm{N}\,]$ summed over contact pts	-10.0
`feet_stumble`	$\mathbb{1}\big[\exists f : \lVert \mathbf{F}_f^{xy}\rVert > 4\lvert F_f^z\rvert\big]$	-1.25
`feet_contact_forces`	$\sum_f \mathrm{ReLU}\big(\lvert F_f^z\rvert - 500\,\mathrm{N}\big)$	-5×10^{-4}
`feet_slip`	$\sum_f \sqrt{\lVert \dot{\mathbf{x}}_f^{xy}\rVert}\,\mathbb{1}[\lvert F_f^z\rvert > 5]$	-0.1
Regularization — smoothness & locomotion		
`dof_vel`	$\sum_j \dot{q}_j^2$	-10^{-4}
`dof_acc`	$\sum_j \ddot{q}_j^2$	-5×10^{-8}
`ankle_dof_vel`	$\sum_{j\in\mathcal{J}_{\mathrm{ank}}} \dot{q}_j^2$	-2×10^{-4}
`ankle_dof_acc`	$\sum_{j\in\mathcal{J}_{\mathrm{ank}}} \ddot{q}_j^2$	-10^{-7}
`action_rate_l2`	$\lVert \mathbf{a}_t - \mathbf{a}_{t-1}\rVert^2$	-0.1
`ang_vel_xy`	$\omega_{b,x}^2 + \omega_{b,y}^2$	-0.01
`feet_air_time`	$\sum_f \min(0, t_f^{\mathrm{air}} - 0.5\mathrm{s})\,\mathbb{1}[\text{first contact}]\,\mathbb{1}[\lVert \mathbf{v}_{b,d}^{xy}\rVert > 0.05]$	$+5.0$

Torque reaction. We report the fraction of near-saturated joints, ρ_τ:

$$\rho_\tau = \tfrac{1}{T n_j} \sum_{t=1}^{T} \sum_{j=1}^{n_j} \mathbf{1}\left[\lvert\tau_{j,t}\rvert > 0.9\tau_j^{\max}\right], \tag{19}$$

where $\tau_j^{\max}$ is the torque limit of joint j. A stiff controller fights the force (high ρ_τ); a compliant one yields, keeping it low.

Lower-body participation. To verify that the controller actually uses the lower body under multi-site contacts, we compare the torque deviation from a matched no-perturbation rollout:

$$R_{\mathrm{LB}} = \frac{\sum_t \sum_{j\in\mathcal{J}_{\mathrm{lower}}} \lvert\tau_{j,t} - \tau_{j,t}^0\rvert}{\sum_t \sum_{j\in\mathcal{J}_{\mathrm{all}}} \lvert\tau_{j,t} - \tau_{j,t}^0\rvert + \epsilon}. \tag{20}$$

Here $\boldsymbol{\tau}^0$ is the torque sequence produced by the same controller on the same command trajectory without external perturbation. A high R_{LB} during hip, pelvis, torso, or knee perturbations indicates that the policy redistributes effort through the legs instead of treating force response as an upper-body-only problem.

F Local passivity by construction

Proposition 1 (Bounded interaction force and local passivity)**.** *Let the closed-loop system satisfy: (i) the base policy realizes $\boldsymbol{\tau}_{\mathrm{c}}$ of Eq. (7) at the equilibrium $(\boldsymbol{q}^\star, \mathbf{0})$; (ii) the residual output is bounded* *as in Eq. (4); (iii) the per-link stiffness $\boldsymbol{K}_\ell$ is constant in time. Then (a) the residual shifts each link equilibrium by at most ϵ_x, so the induced restoring force is bounded by $\|\boldsymbol{K}_\ell\|\,\epsilon_x$; and (b) if additionally (iv) the residual-edited equilibrium $\tilde{\boldsymbol{x}}_{\ell,d}$ varies slowly enough that its power injection is dominated by the link damping $\boldsymbol{D}_\ell$, the storage function $S = \frac{1}{2}\boldsymbol{\nu}^\top \boldsymbol{M}\boldsymbol{\nu} + \frac{1}{2}\sum_\ell (\boldsymbol{x}_\ell - \tilde{\boldsymbol{x}}_{\ell,d})^\top \boldsymbol{K}_\ell (\boldsymbol{x}_\ell - \tilde{\boldsymbol{x}}_{\ell,d})$ satisfies $\dot{S} \le \sum_i \boldsymbol{f}_{ext,i}^\top \dot{\boldsymbol{x}}_{ext,i}$, where $\boldsymbol{x}_{ext,i}$ is the displacement of contact point i; i.e. the closed loop is passive with respect to the external-wrench port.*

Sketch. Bound (a) is immediate from Eq. (4): the residual moves the equilibrium by at most ϵ_x, so with fixed $\boldsymbol{K}_\ell$ the induced restoring force changes by at most $\|\boldsymbol{K}_\ell\|\,\epsilon_x$. For (b), differentiating S and substituting the closed-loop dynamics gives $\dot{S} = \sum_i \boldsymbol{f}_{\text{ext},i}^\top \dot{\boldsymbol{x}}_{\text{ext},i} - (\text{damping dissipation}) - \sum_\ell (\boldsymbol{x}_\ell - \tilde{\boldsymbol{x}}_{\ell,d})^\top \boldsymbol{K}_\ell\, \dot{\tilde{\boldsymbol{x}}}_{\ell,d}$. With $\boldsymbol{K}_\ell$ constant the $\dot{\boldsymbol{K}}_\ell$ term that would otherwise inject non-passive power vanishes identically; the remaining moving-equilibrium term is dominated by the damping dissipation under assumption (iv), yielding the stated inequality. For the general time-varying-stiffness case the energy tank of §A would absorb the $\dot{\boldsymbol{K}}_\ell$ term; in our pipeline stiffness is fixed and the residual does not modulate gains, so the tank is reference scaffolding rather than a runtime mechanism.

Scope and what this does and does not buy. The result is *local* (around the equilibrium) and *by construction*: it follows from the residual bound and fixed stiffness, not from the learned components. Global stability under adversarial external forces is not claimed and remains open. Two consequences are worth stating explicitly. (i) Action-residual baselines (ASAP, ResMimic) edit the torque directly and do not admit this storage function; their effect on induced contact force is not physically bounded by construction. (ii) The proposition is a safety property for human–robot contact, not a performance property: it does not guarantee tracking, success rate, or sim-to-real transfer. Those are settled empirically (§5).